MySQL Professional Student Assessment

Project: Student Management System Database

Part 1: Database Design

1 Create a database named student_management

```sql
CREATE DATABASE student_management;
USE student_management;
```

2 Create tables: Students, Courses, Enrollments, Marks & 3 Apply constraints: Primary Key, Foreign Key, Unique, Not Null, Default

```sql
CREATE TABLE Students (
    student_id INT PRIMARY KEY,
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL,
    email VARCHAR(100) UNIQUE,
    enrollment_date DATE DEFAULT '2024-01-01'
);

CREATE TABLE Courses (
    course_id INT PRIMARY KEY,
    course_name VARCHAR(100) NOT NULL,
    credits INT DEFAULT 3
);

CREATE TABLE Enrollments (
    enrollment_id INT PRIMARY KEY,
    student_id INT,
    course_id INT,
    FOREIGN KEY (student_id) REFERENCES Students(student_id),
    FOREIGN KEY (course_id) REFERENCES Courses(course_id)
);

CREATE TABLE Marks (
    mark_id INT PRIMARY KEY,
    enrollment_id INT,
    score INT NOT NULL,
    total_marks INT DEFAULT 100,
    FOREIGN KEY (enrollment_id) REFERENCES Enrollments(enrollment_id)
);
```

Part 2: Data Insertion

Insert sample data: minimum 10 students, 5 courses, 15 enrollments, and 20 marks.

Run SQL query/queries on database student_management:

```sql
INSERT INTO Students (student_id, first_name, last_name, email, enrollment_date) VALUES
(1, 'Amit', 'Sharma', 'amit@email.com', '2024-01-10'),
(2, 'Suman', 'Patil', 'suman@email.com', '2024-01-11'),
(3, 'Rahul', 'Verma', 'rahul@email.com', '2024-01-12'),
(4, 'Ananya', 'Das', 'ananya@email.com', '2024-01-13'),
(5, 'Vijay', 'Kumar', 'vijay@email.com', '2024-01-14'),
(6, 'Priya', 'Singh', 'priya@email.com', '2024-01-15'),
(7, 'Arjun', 'Reddy', 'arjun@email.com', '2024-01-16'),
(8, 'Sneha', 'Nair', 'sneha@email.com', '2024-01-17'),
(9, 'Rohan', 'Mehta', 'rohan@email.com', '2024-01-18'),
(10, 'Kavya', 'Gowda', 'kavya@email.com', '2024-01-19');
```

Run SQL query/queries on database student_management:

```sql
INSERT INTO Courses (course_id, course_name, credits) VALUES
(101, 'Mathematics', 4),
(102, 'Physics', 4),
(103, 'Computer Science', 3),
(104, 'English', 2),
(105, 'History', 2);
```

Run SQL query/queries on database student_management:

```sql
INSERT INTO Enrollments (enrollment_id, student_id, course_id) VALUES
(1, 1, 101), (2, 1, 102), (3, 2, 101), (4, 3, 103), (5, 4, 101),
(6, 4, 104), (7, 5, 102), (8, 6, 105), (9, 7, 103), (10, 8, 101),
(11, 8, 102), (12, 9, 104), (13, 10, 103), (14, 10, 105), (15, 2, 103);
```

Run SQL query/queries on database student_management:

```sql
INSERT INTO Marks (mark_id, enrollment_id, score) VALUES
(1, 1, 85), (2, 2, 78), (3, 3, 92), (4, 4, 88), (5, 5, 76),
(6, 6, 95), (7, 7, 82), (8, 8, 70), (9, 9, 89), (10, 10, 91),
(11, 11, 84), (12, 12, 77), (13, 13, 90), (14, 14, 83), (15, 15, 86),
(16, 1, 90), (17, 3, 80), (18, 5, 88), (19, 10, 75), (20, 13, 94);
```

Part 3: SQL Queries

1 Basic queries: SELECT, WHERE, ORDER BY, LIMIT

1.SELECT first_name, last_name FROM Students;

```sql
SELECT * FROM Students;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	student_id	first_name	last_name	email	enrollment_date
Edit Copy Delete	1	Amit	Sharma	amit@email.com	2024-01-10
Edit Copy Delete	2	Suman	Patil	suman@email.com	2024-01-11
Edit Copy Delete	3	Rahul	Verma	rahul@email.com	2024-01-12
Edit Copy Delete	4	Ananya	Das	ananya@email.com	2024-01-13
Edit Copy Delete	5	Vijay	Kumar	vijay@email.com	2024-01-14
Edit Copy Delete	6	Priya	Singh	priya@email.com	2024-01-15
Edit Copy Delete	7	Arjun	Reddy	arjun@email.com	2024-01-16
Edit Copy Delete	8	Sneha	Nair	sneha@email.com	2024-01-17
Edit Copy Delete	9	Rohan	Mehta	rohan@email.com	2024-01-18
Edit Copy Delete	10	Kavya	Gowda	kavya@email.com	2024-01-19

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2.SELECT * FROM Students WHERE enrollment_date > '2024-01-12';

```
SELECT * FROM Students WHERE enrollment_date > '2024-01-12';
```

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Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	student_id	first_name	last_name	email	enrollment_date
Edit Copy Delete	4	Ananya	Das	ananya@email.com	2024-01-13
Edit Copy Delete	5	Vijay	Kumar	vijay@email.com	2024-01-14
Edit Copy Delete	6	Priya	Singh	priya@email.com	2024-01-15
Edit Copy Delete	7	Arjun	Reddy	arjun@email.com	2024-01-16
Edit Copy Delete	8	Sneha	Nair	sneha@email.com	2024-01-17
Edit Copy Delete	9	Rohan	Mehta	rohan@email.com	2024-01-18
Edit Copy Delete	10	Kavya	Gowda	kavya@email.com	2024-01-19

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3.SELECT * FROM Students ORDER BY last_name ASC;

```
SELECT * FROM Students ORDER BY last_name ASC;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refre

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	student_id	first_name	last_name 1	email	enrollment_date
Edit Copy Delete	4	Ananya	Das	ananya@email.com	2024-01-13
Edit Copy Delete	10	Kavya	Gowda	kavya@email.com	2024-01-19
Edit Copy Delete	5	Vijay	Kumar	vijay@email.com	2024-01-14
Edit Copy Delete	9	Rohan	Mehta	rohan@email.com	2024-01-18
Edit Copy Delete	8	Sneha	Nair	sneha@email.com	2024-01-17
Edit Copy Delete	2	Suman	Patil	suman@email.com	2024-01-11
Edit Copy Delete	7	Arjun	Reddy	arjun@email.com	2024-01-16
Edit Copy Delete	1	Amit	Sharma	amit@email.com	2024-01-10
Edit Copy Delete	6	Priya	Singh	priya@email.com	2024-01-15
Edit Copy Delete	3	Rahul	Verma	rahul@email.com	2024-01-12

4.SELECT * FROM Students LIMIT 5;

```
SELECT * FROM Students LIMIT 5;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [R

Extra options

	student_id	first_name	last_name	email	enrollment_date
Edit Copy Delete	1	Amit	Sharma	amit@email.com	2024-01-10
Edit Copy Delete	2	Suman	Patil	suman@email.com	2024-01-11
Edit Copy Delete	3	Rahul	Verma	rahul@email.com	2024-01-12
Edit Copy Delete	4	Ananya	Das	ananya@email.com	2024-01-13
Edit Copy Delete	5	Vijay	Kumar	vijay@email.com	2024-01-14

2 Logical operators: AND, OR, NOT

a) SELECT * FROM Students WHERE first_name = 'Amit' AND last_name = 'Sharma';

```
SELECT * FROM Students WHERE first_name = 'Amit' AND last_name = 'Sharma';
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

	student_id	first_name	last_name	email	enrollment_date
Edit Copy Delete	1	Amit	Sharma	amit@email.com	2024-01-10

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b) SELECT * FROM Students WHERE first_name = 'Amit' OR first_name = 'Priya';

```
SELECT * FROM Students WHERE first_name = 'Amit' OR first_name = 'Priya';
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refre

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	student_id	first_name	last_name	email	enrollment_date
Edit Copy Delete	1	Amit	Sharma	amit@email.com	2024-01-10
Edit Copy Delete	6	Priya	Singh	priya@email.com	2024-01-15

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c) SELECT * FROM Students WHERE NOT first_name = 'Rahul';

```
SELECT * FROM Students WHERE NOT first_name = 'Rahul';
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refr

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	student_id	first_name	last_name	email	enrollment_date
Edit Copy Delete	1	Amit	Sharma	amit@email.com	2024-01-10
Edit Copy Delete	2	Suman	Patil	suman@email.com	2024-01-11
Edit Copy Delete	4	Ananya	Das	ananya@email.com	2024-01-13
Edit Copy Delete	5	Vijay	Kumar	vijay@email.com	2024-01-14
Edit Copy Delete	6	Priya	Singh	priya@email.com	2024-01-15
Edit Copy Delete	7	Arjun	Reddy	arjun@email.com	2024-01-16
Edit Copy Delete	8	Sneha	Nair	sneha@email.com	2024-01-17
Edit Copy Delete	9	Rohan	Mehta	rohan@email.com	2024-01-18
Edit Copy Delete	10	Kavya	Gowda	kavya@email.com	2024-01-19

3 Aggregate functions: MIN, MAX, COUNT, AVG, SUM

SELECT MIN(score) FROM Marks;

MIN(score)
70

SELECT MAX(score) FROM Marks;

MAX(score)
95

SELECT COUNT(*) FROM Students;

COUNT(*)
10

SELECT AVG(score) FROM Marks;

AVG(score)
84.6500

SELECT SUM(credits) FROM Courses;

SUM(credits)
15

4 Filtering: LIKE, IN, BETWEEN

SELECT * FROM Students WHERE first_name LIKE 'A%';

	student_id	first_name	last_name	email	enrollment_date
Edit Copy Delete	1	Amit	Sharma	amit@email.com	2024-01-10
Edit Copy Delete	4	Ananya	Das	ananya@email.com	2024-01-13
Edit Copy Delete	7	Arjun	Reddy	arjun@email.com	2024-01-16

SELECT * FROM Students WHERE student_id IN (1, 3, 5);

	student_id	first_name	last_name	email	enrollment_date
Edit Copy Delete	1	Amit	Sharma	amit@email.com	2024-01-10
Edit Copy Delete	3	Rahul	Verma	rahul@email.com	2024-01-12
Edit Copy Delete	5	Vijay	Kumar	vijay@email.com	2024-01-14

SELECT * FROM Marks WHERE score BETWEEN 80 AND 90;

	mark_id	enrollment_id	score	total_marks
Edit Copy Delete	1	1	85	100
Edit Copy Delete	4	4	88	100
Edit Copy Delete	7	7	82	100
Edit Copy Delete	9	9	89	100
Edit Copy Delete	11	11	84	100
Edit Copy Delete	13	13	90	100
Edit Copy Delete	14	14	83	100
Edit Copy Delete	15	15	86	100
Edit Copy Delete	16	1	90	100
Edit Copy Delete	17	3	80	100
Edit Copy Delete	18	5	88	100

5 Joins: INNER, LEFT, RIGHT, CROSS, SELF

a) SELECT Students.first_name, Enrollments.enrollment_id
FROM Students
INNER JOIN Enrollments ON Students.student_id = Enrollments.student_id;

first_name	enrollment_id
Amit	1
Amit	2
Suman	3
Suman	15
Rahul	4
Ananya	5
Ananya	6
Vijay	7
Priya	8
Arjun	9
Sneha	10
Sneha	11
Rohan	12
Kavya	13
Kavya	14

b) SELECT Students.first_name, Enrollments.enrollment_id
FROM Students
LEFT JOIN Enrollments ON Students.student_id = Enrollments.student_id;

first_name	enrollment_id
Amit	1
Amit	2
Suman	3
Suman	15
Rahul	4
Ananya	5
Ananya	6
Vijay	7
Priya	8
Arjun	9
Sneha	10
Sneha	11
Rohan	12
Kavya	13
Kavya	14

c) SELECT Students.first_name, Enrollments.enrollment_id
FROM Students
RIGHT JOIN Enrollments ON Students.student_id = Enrollments.student_id;

first_name	enrollment_id
Amit	1
Amit	2
Suman	3
Suman	15
Rahul	4
Ananya	5
Ananya	6
Vijay	7
Priya	8
Arjun	9
Sneha	10
Sneha	11
Rohan	12
Kavya	13
Kavya	14

d) SELECT Students.first_name, Courses.course_name
FROM Students
CROSS JOIN Courses;

first_name	course_name
Amit	Mathematics
Amit	Physics
Amit	Computer Science
Amit	English
Amit	History
Suman	Mathematics
Suman	Physics
Suman	Computer Science
Suman	English
Suman	History
Rahul	Mathematics
Rahul	Physics
Rahul	Computer Science
Rahul	English
Rahul	History
Ananya	Mathematics
Ananya	Physics
Ananya	Computer Science
Ananya	English
Ananya	History
Vijay	Mathematics
Vijay	Physics
Vijay	Computer Science
Vijay	English
Vijay	History

e) SELECT A.first_name AS Student1, B.first_name AS Student2
FROM Students A, Students B
WHERE A.student_id <> B.student_id;

Student1	Student2
Rahul	Amit
Ananya	Amit
Vijay	Amit
Priya	Amit
Arjun	Amit
Sneha	Amit
Rohan	Amit
Kavya	Amit
Amit	Suman
Rahul	Suman
Ananya	Suman
Vijay	Suman
Priya	Suman
Arjun	Suman
Sneha	Suman
Rohan	Suman
Kavya	Suman
Amit	Rahul
Suman	Rahul
Ananya	Rahul
Vijay	Rahul
Priya	Rahul
Arjun	Rahul
Sneha	Rahul

6 Grouping: GROUP BY, HAVING

a) SELECT course_id, COUNT(student_id)
FROM Enrollments
GROUP BY course_id;

course_id	COUNT(student_id)
101	4
102	3
103	4
104	2
105	2

b) SELECT course_id, COUNT(student_id)
FROM Enrollments
GROUP BY course_id HAVING COUNT(student_id) > 2;

course_id	COUNT(student_id)
101	4
102	3
103	4

7 Advanced: UNION, EXISTS, CASE

a)

```
SELECT first_name FROM Students UNION SELECT course_name FROM Courses;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

first_name
Amit
Suman
Rahul
Ananya
Vijay
Priya
Arjun
Sneha
Rohan
Kavya
Mathematics
Physics
Computer Science
English
History

b)

```
SELECT course_name FROM Courses WHERE EXISTS (SELECT * FROM Enrollments WHERE Enrollments.course_id =
Courses.course_id);
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	course_name
Edit Copy Delete	Mathematics
Edit Copy Delete	Physics
Edit Copy Delete	Computer Science
Edit Copy Delete	English
Edit Copy Delete	History

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c)

```
SELECT score, CASE WHEN score >= 90 THEN 'Excellent' WHEN score >= 80 THEN 'Very Good' ELSE 'Good' END AS Grade FROM Marks;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	score	Grade
Edit Copy Delete	85	Very Good
Edit Copy Delete	78	Good
Edit Copy Delete	92	Excellent
Edit Copy Delete	88	Very Good
Edit Copy Delete	76	Good
Edit Copy Delete	95	Excellent
Edit Copy Delete	82	Very Good
Edit Copy Delete	70	Good
Edit Copy Delete	89	Very Good
Edit Copy Delete	91	Excellent
Edit Copy Delete	84	Very Good
Edit Copy Delete	77	Good
Edit Copy Delete	90	Excellent
Edit Copy Delete	83	Very Good
Edit Copy Delete	86	Very Good
Edit Copy Delete	90	Excellent
Edit Copy Delete	80	Very Good
Edit Copy Delete	88	Very Good
Edit Copy Delete	75	Good
Edit Copy Delete	94	Excellent

Part 4: Data Modification

Perform UPDATE, DELETE, and INSERT SELECT operations.

a) UPDATE Students SET email = 'amit_new@email.com' WHERE student_id = 1;

Before:

←T→	student_id	first_name	last_name	email	enrollment_date
Edit Copy Delete	1	Amit	Sharma	amit@email.com	2024-01-10

After:

Edit Copy Delete	1	Amit	Sharma	amit_new@email.com	2024-01-10

b) DELETE FROM Marks WHERE score < 40;

Before:

Edit Copy Delete	21	1	35	100

After:

Edit Copy Delete	2	2	78	100
Edit Copy Delete	3	3	92	100
Edit Copy Delete	4	4	88	100
Edit Copy Delete	5	5	76	100
Edit Copy Delete	6	6	95	100
Edit Copy Delete	7	7	82	100
Edit Copy Delete	8	8	70	100
Edit Copy Delete	9	9	89	100
Edit Copy Delete	10	10	91	100
Edit Copy Delete	11	11	84	100
Edit Copy Delete	12	12	77	100
Edit Copy Delete	13	13	90	100
Edit Copy Delete	14	14	83	100
Edit Copy Delete	15	15	86	100
Edit Copy Delete	16	1	90	100
Edit Copy Delete	17	3	80	100
Edit Copy Delete	18	5	88	100
Edit Copy Delete	19	10	75	100
Edit Copy Delete	20	13	94	100

c) INSERT INTO Enrollments (enrollment_id, student_id, course_id)

SELECT 16, student_id, 101

FROM Students

WHERE student_id = 5;

Before:

	enrollment_id	student_id	course_id
Edit Copy Delete	1	1	101
Edit Copy Delete	2	1	102
Edit Copy Delete	3	2	101
Edit Copy Delete	4	3	103
Edit Copy Delete	5	4	101
Edit Copy Delete	6	4	104
Edit Copy Delete	7	5	102

After:

Edit Copy Delete	16	5	101

Part 5: Advanced Features

1 Null handling functions (IFNULL, COALESCE)

2 Views creation

```
CREATE VIEW StudentGrades AS
 SELECT Students.first_name, Courses.course_name, Marks.score
 FROM Students
JOIN Enrollments ON Students.student_id = Enrollments.student_id
JOIN Courses ON Enrollments.course_id = Courses.course_id
JOIN Marks ON Enrollments.enrollment_id = Marks.enrollment_id;
```

SELECT * FROM StudentGrades;

Table	Action	Rows	Type	Collation	Size	Over
studentgrades	Browse Structure Search Insert Edit Drop	~0	View	---	-	
1 table	Sum	~0	InnoDB	utf8mb4_general_ci	0 B	

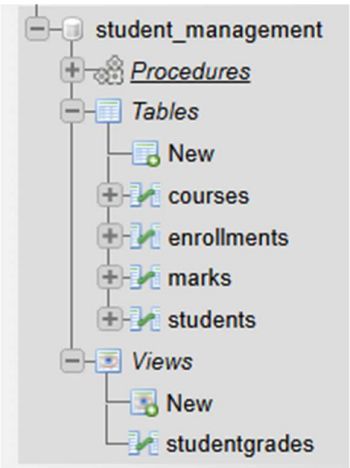
Check all With selected:

Print Data dictionary

first_name	course_name	score
Amit	Mathematics	85
Amit	Mathematics	90
Suman	Mathematics	92
Suman	Mathematics	80
Ananya	Mathematics	76
Ananya	Mathematics	88
Sneha	Mathematics	91
Sneha	Mathematics	75
Amit	Physics	78
Vijay	Physics	82
Sneha	Physics	84
Rahul	Computer Science	88
Arjun	Computer Science	89
Kavya	Computer Science	90
Kavya	Computer Science	94
Suman	Computer Science	86
Ananya	English	95
Rohan	English	77
Priya	History	70
Kavya	History	83

3 Stored Procedures

```
DELIMITER //

 CREATE PROCEDURE GetAllStudents()

BEGIN

        SELECT * FROM Students;

END //

DELIMITER ;

CALL GetAllStudents();
```



	Name	Type	Returns	
☐	GetAllStudents	PROCEDURE		Edit Execute Export Drop